

# DAY 1 – TEAM 7 PROGRESS REPORT

## *Summary of Team Activities*

| Team          | Day 1 Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware Team | <p>For this design , we decided to use “ATMEGA8A-PU” <a href="https://makerselectronics.com/product/atmega8a-pu-8-bit-avr-microcontroller-dip-28-original/">https://makerselectronics.com/product/atmega8a-pu-8-bit-avr-microcontroller-dip-28-original/</a> as our microcontroller , its an 8-bit AVR microcontroller with 8KB Flash, 23 I/O pins, 16MHz clock. Supports USART, SPI, I2C. paired with an FTTL , it will be easy to program using arduino IDE</p> <p>For the current sensor , we decided to use <a href="https://makerselectronics.com/product/current-sensor-module-ac712-5a/">https://makerselectronics.com/product/current-sensor-module-ac712-5a/</a> , very easy to use , and its got its own library in arduino IDE for easier software , and satisfies the 5A max requirement ( motors alone could pull at least 4A )</p> <p>For the Voltmeter module , we decided to use <a href="https://makerselectronics.com/product/digital-voltmeter-display-red-led-0/">https://makerselectronics.com/product/digital-voltmeter-display-red-led-0/</a> , its a Compact 0.56" red-LED digital voltmeter (2-wire, 4.5–30 VDC) self-powered, easy-install panel meter , batteries combined would not reach 30V so we are safe at this level</p> |
| Software Team | Reviewed the available resources and gathered relevant information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Firmware Team | Conducted research on each assigned task and created a diagram illustrating the overall system integration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |